Microeconomics

Lecture 3 · Duality, Slutsky and revealed preference

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What does a price change actually do?

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- Connect the two consumer problems: duality and Shephard’s lemma.
- Separate substitution from purchasing power: the Slutsky equation.
- Test the model: demand derivatives and observed choices.

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Today: fix utility instead of money.

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Two uses: isolate substitution; measure compensation in money.

The target is ordinal: a bundle's rank matters, not the numerical utility scale.

The expenditure minimisation problem

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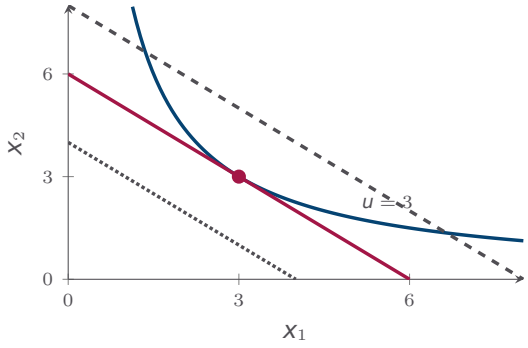
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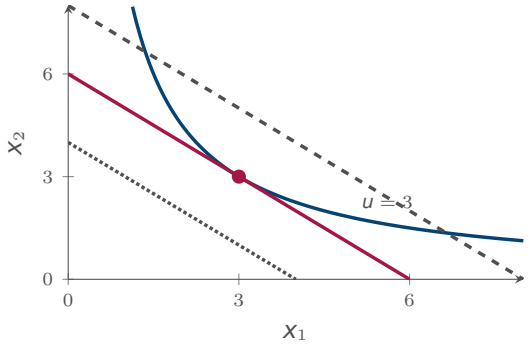
e: expenditure function. H: Hicksian demand correspondence; h if unique.

Move the budget line towards the origin



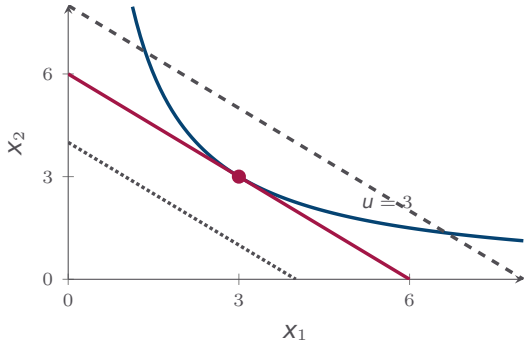
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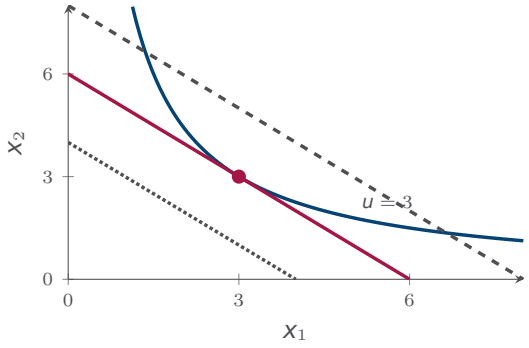
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Cost 4 cannot attain it.

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- Add locally nonsatiated preferences for the duality results.
- Add uniqueness and smoothness only where derivatives require them.

Does a minimum exist?

Proposition. If u is continuous, $p \gg 0$ and $\bar{u}$ is attainable, then $H(p, \bar{u}) \neq \varnothing$.

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How can we still use a compactness argument?

Find *one* feasible bundle and discard every bundle that costs more.

Existence: cut away expensive bundles

Choose $\widehat{x} \geq 0$ with $u(\widehat{x}) \geq \bar{u}$. Set

$$K = \{x \geq 0 : u(x) \geq \bar{u},\ p \cdot x \leq p \cdot \widehat{x}\}.$$

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Every excluded feasible bundle costs more than $p \cdot \widehat{x}$, so this is a global minimum.

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Otherwise, continuity gives $t < 1$ close to 1 with $u(th) > \bar{u}$.

But $p \cdot (th) < p \cdot h$: a cheaper feasible bundle. Contradiction.

Change the currency, not the chosen bundle

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Hicksian demand has degree 0 in prices; expenditure has degree 1.

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Two distinct minimisers have utility exactly $\bar{u}$. Their mixture would give more than $\bar{u}$, violating no excess utility.

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Otherwise a minimiser for $\bar{u}'$ would also minimise cost for $\bar{u}$, contradicting no excess utility.

Expenditure is concave in prices

Proposition. For a fixed attainable target and $0 \le t \le 1$,

$$e(tp + (1-t)q, \bar{u}) \ge t\,e(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

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The same bundle must be purchased at the mixed prices. Reoptimising separately can only be cheaper.

Concavity: one bundle, two price vectors

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Therefore

$$\begin{aligned} e(tp + (1-t)q, \bar{u}) &= (tp + (1-t)q) \cdot z \\ &\geq t\, e(p, \bar{u}) + (1-t) e(q, \bar{u}). \end{aligned}$$

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With continuous LNS utility, e is continuous in positive prices and interior attainable targets.

Buy a Cobb–Douglas standard of living

NES

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At $p = (4, 1)$ and $\bar{u} = 3$, what are the quantities and minimum cost?

Cobb–Douglas: the dual solution

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At $p = (4, 1)$ and $\bar{u} = 3$: $h = (3/2, 6)$, $e = 12$.

Compare $x_i(p, m) = m/(2p_i)$: Hicksian demand fixes $\bar{u}$, not m.

The kink is easy: perfect complements

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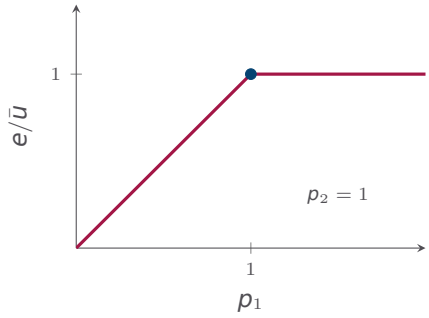
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Can the direct utility have a kink while expenditure is smooth in prices?

Yes. Here expenditure is linear in prices and compensated quantities do not change.

The cheapest route can switch

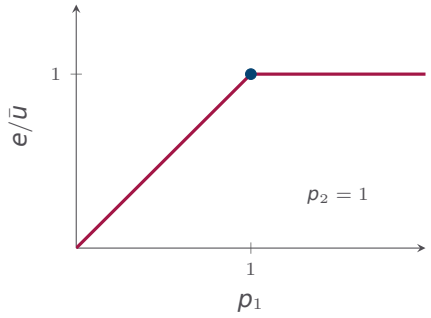
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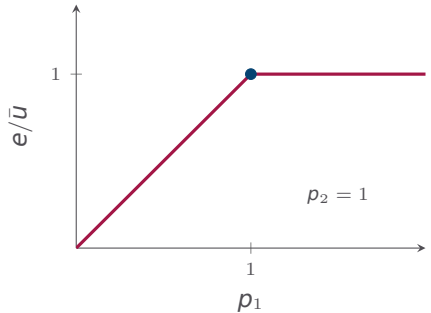


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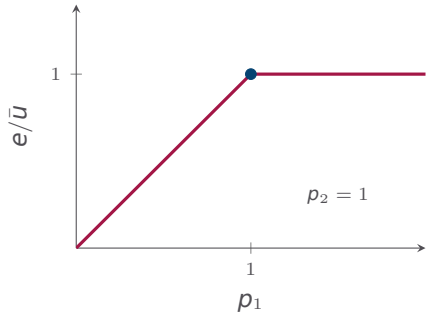


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$p_1 < p_2$: $h = (\bar{u}, 0)$.
$p_1 > p_2$: $h = (0, \bar{u})$.
$p_1 = p_2$: every $x \geq 0$ with $x_1 + x_2 = \bar{u}$ minimises cost.



Two problems, the same boundary

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The equalities hold for *sets of choices*, including nonunique optima.

Proof: the cheapest way to buy the maximum

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If $e(p, \bar{u}) < m$, a minimiser h attains at least $\bar{u}$ at cost below m.

Then h is also utility-maximising with a slack budget, contradicting LNS and Walras. Thus $e(p, v(p, m)) = m$.

Proof: the most utility the minimum cost buys

Choose $h \in H(p, \bar{u})$ and set $m = e(p, \bar{u})$. No excess utility gives $u(h) = \bar{u}$.

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That violates no excess utility. Hence $v(p, e(p, \bar{u})) = \bar{u}$.

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Substitute $m = e(p, \bar{u})$ for the other equality.

Check the identities by hand

For $u = \sqrt{x_1 x_2}$:

$$v(p, m) = \frac{m}{2\sqrt{p_1 p_2}}, \qquad e(p, \bar{u}) = 2\bar{u}\sqrt{p_1 p_2}.$$

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Same choices at matching parameters; different responses when prices change.

Shephard’s lemma

Fix an attainable target and positive prices. If the Hicksian minimiser is unique, expenditure is differentiable in prices and

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

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Equivalently, wherever e is differentiable, every minimiser equals $\nabla_p e$.

Interpretation: a small price increase costs the compensated consumer approximately $h_i \, dp_i$.

Why the quantity appears in the derivative

Let $h = h(p, \bar{u})$ and choose $h_t \in H(p + te_i, \bar{u})$, where e_i is the ith unit vector.

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Divide by t and take limits; negative t gives the same derivative h_i.

What if demand is not unique?

For any $h \in H(p, \bar{u})$ and any $q \gg 0$,

$$e(q, \bar{u}) \leq q \cdot h = e(p, \bar{u}) + (q - p) \cdot h.$$

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Use the demand correspondence at a kink; a single derivative cannot select the whole set.

Recover quantities from cost

Cobb–Douglas: $e = 2\bar{u}\sqrt{p_1 p_2}$.

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Substitutes: $e = \bar{u}\min\{p_1, p_2\}$; differentiate only within a strict price regime.

The compensated law of demand

Fix $\bar{u}$. Let $h \in H(p, \bar{u})$ and $k \in H(q, \bar{u})$. Then

$$(q - p) \cdot (k - h) \leq 0.$$

Why is it essential to hold the target fixed?

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No differentiability, uniqueness or convex preferences are required.

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For a differentiable unique demand: $\partial h_i / \partial p_i \leq 0$.

The matrix behind compensated substitution

Assume $e(\cdot, \bar{u})$ is C^2 locally. Define

$$S_{ij} = \frac{\partial h_i}{\partial p_j}, \qquad S = D_p h = D_{pp}^2 e.$$

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In particular, $S_{ii} \leq 0$. Zero effects are allowed.

Why symmetry and the quadratic inequality?

By Shephard, $h_i = e_{p_i}$. Mixed partials of a C^2 function commute:

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Concavity of e implies concavity of g, hence

$$g''(0) = z^\top D^2_{pp} e(p, \bar{u}) z \leq 0.$$

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Recall $h(tp, \bar{u}) = h(p, \bar{u})$. Differentiate in t at $t = 1$:

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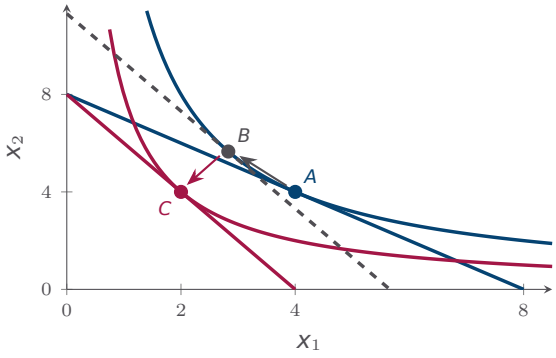
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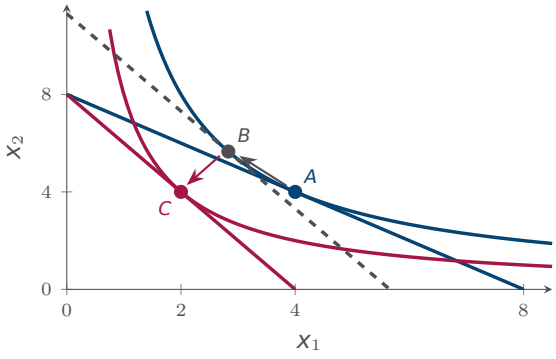
Relative prices drive compensated substitution; a change of currency does not.

A price increase: three choices on one picture



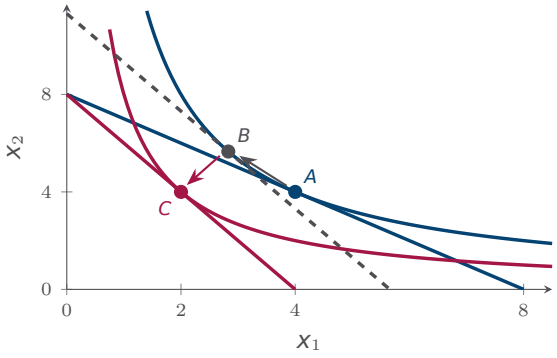
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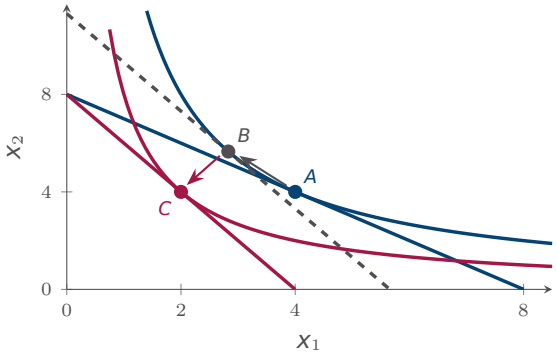
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Keep $u^0 = 4$:
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$p^1 = (2, 1)$: $C = (2, 4)$.
Keep $u^0 = 4$:
$B = (2\sqrt{2}, 4\sqrt{2})$.
$A \to B$: substitution.
$B \to C$: income effect.

The choices match. Their price derivatives do not.

$$h(p, \bar{u}) = x(p, e(p, \bar{u})).$$

Which two terms appear when we differentiate the right-hand side?

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On the left, change prices while holding $\bar{u}$ fixed.

On the right, change prices *and adjust income* to $e(p, \bar{u})$.

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Slutsky: the chain rule does the work

Assume unique demands with the derivatives below. Start from

$$h_i(p, \bar{u}) = x_i(p, e(p, \bar{u})).$$

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Shephard gives $e_{p_j} = h_j$. At $\bar{u} = v(p, m)$, $h_j = x_j$ and $e = m$.

The Slutsky equation

At $\bar{u} = v(p, m)$:

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total price effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{purchasing-power term}}.$$

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The quantity multiplying the income derivative is x_j, not x_i.

When must ordinary demand slope down?

For the own price:

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Giffen good: $\partial x_i / \partial p_i > 0$; the positive income effect dominates.

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At $p = (1, 1)$, $m = 8$: total -4 = substitution -2 + income -2.

No total effect does not mean no substitution

Cobb–Douglas gives $x_1 = m/(2p_1)$, so $\partial x_1/\partial p_2 = 0$.

Does good 2 have no effect on the demand for good 1?

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The positive compensated substitution effect and negative income effect cancel exactly.

At $p = (1, 1)$, $m = 8$: $0 = 2 - 2$.

Perfect complements: all income, no substitution

For $u = \min\{x_1, x_2\}$:

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For either i, j,

$$\frac{\partial x_i}{\partial p_j} = -\frac{m}{(p_1 + p_2)^2} = 0 - \frac{m}{p_1 + p_2}\frac{1}{p_1 + p_2}.$$

Perfect complements: all income, no substitution

For $u = \min\{x_1, x_2\}$:

$$x_1 = x_2 = \frac{m}{p_1 + p_2}, \qquad h_1 = h_2 = \bar{u}.$$

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Ordinary demands fall together even though compensated quantities stay fixed.

Quasilinearity: income effects depend on the regime

$u(a, q) = a + 4\log(1 + q)$, $a + rq \leq m$, $a, q \geq 0$.

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NES

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“No income effect” applies within the interior numeraire regime.

Can a higher price make you buy more?

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At fixed tastes and income, a Giffen response means $\partial x_i / \partial p_i > 0$.

Can we construct a Giffen good with continuous, monotone, convex preferences?

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Yes. A smooth utility function is not needed for a smooth local demand.

A subsistence kink generates Giffen demand

Let

$$u(x) = \min\{x_1 + x_2,\ 3x_1 + 2x_2 - 10\}.$$

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$$x_1 = \frac{10p_2 - m}{2p_2 - p_1}, \qquad x_2 = \frac{2m - 10p_1}{2p_2 - p_1}.$$

The utility is continuous, strongly increasing and concave.

The income effect wins

Fix $p_2 = 1$, $m = 8$. When $p_1 = 1.2$, $x = (2.5, 5)$.

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At the original point, $h_{1,p_1} = 0$ and Slutsky gives $3.125 = 0 - (2.5)(-1.25)$.

Substitutes and complements: specify the experiment

For $i \neq j$, **net** substitution uses $\partial h_i / \partial p_j$: positive means net substitutes; negative means net complements.

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With three or more goods, negative net cross-effects are possible.

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Test: $S = S^\top$, $z^\top S z \leq 0$, $Sp = 0$, plus homogeneity and Walras.

Cobb–Douglas passes the matrix test

For $x_i = m/(2p_i)$,

$$S = \frac{m}{4}\begin{pmatrix} -1/p_1^2 & 1/(p_1p_2) \\ 1/(p_1p_2) & -1/p_2^2 \end{pmatrix}.$$

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At $p = (1,1)$, $m = 8$: $S = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$.

Could these choices come from our consumer model?

NES

Suppose estimated demand is

$$x_1(p, m) = \frac{m p_1}{p_1^2 + p_2^2}, \qquad x_2(p, m) = \frac{m p_2}{p_1^2 + p_2^2}.$$

At $p = (1, 1)$ and $m = 2$, calculate the Slutsky matrix. Does the model survive?

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Start with $x = (1, 1)$ and $D_m x = (1/2, 1/2)^\top$.

A symmetric matrix can still fail

At $p = (1, 1)$, $m = 2$,

$$D_p x = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

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The demand violates compensated demand restrictions. It cannot be generated by the maintained consumer model.

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Use affordability comparisons to look for inconsistent choices.

Direct revealed preference

Observation t directly reveals x^t weakly preferred to x^s if

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The reverse affordability comparison is strict, so indifference cannot explain it.

Two shopping trips

Observation	Prices	Chosen bundle	Spending
A	$(2, 1)$	$(2, 1)$	5
B	$(1, 2)$	$(1, 2)$	5

Can one stable, continuous LNS preference relation rationalise both choices?

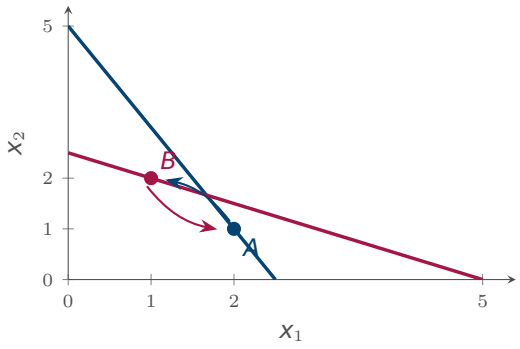
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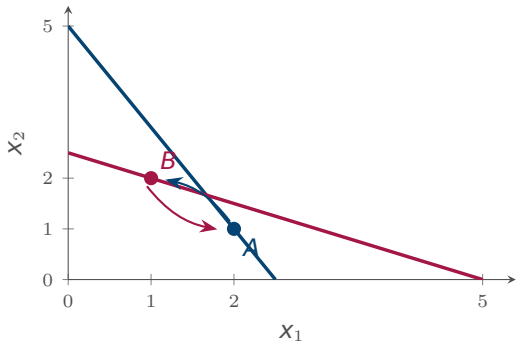
What does bundle B cost at prices A? What does bundle A cost at prices B?

Both rejected bundles were strictly cheaper



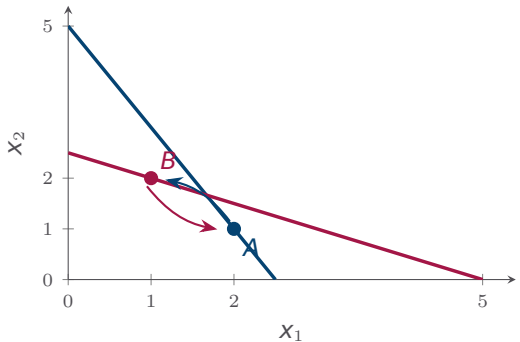
Blue: $2x_1 + x_2 = 5$.
Red: $x_1 + 2x_2 = 5$.

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WARP: a test, not a diagnosis

Weak Axiom of Revealed Preference: exclude a two-observation reversal with at least one strict affordability inequality.

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A rejection concerns the joint model: stable preferences, the recorded goods, prices and feasible sets.

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Passing this pairwise test does not rule out longer revealed-preference cycles.

Checkpoint

- Strictly convex preferences imply strictly concave expenditure in prices.

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- A compensated own-price effect must be strictly negative.
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- A Giffen good with positive consumption must be inferior at that point.

One structure, several ways to use it

$$u \longrightarrow v,\ x \qquad \longleftrightarrow \qquad e,\ h.$$

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Slutsky: connect the two responses and obtain restrictions we can test.

Read next

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Chapter 7, *Revealed Preference*; Chapter 8, *Slutsky Equation*.

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Homework 1 covers Lectures 1–2 and is due **14 November 2026**.

Reserve: compensation and equivalent variation

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CV restores old utility at new prices; EV removes the equivalent income at old prices.

Reserve: how much compensation?

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$$u^0 = 4, \qquad u^1 = 2\sqrt{2}.$$

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These are exact finite-change measures; a derivative is only a local approximation.

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To first order they agree: $de = h(p^0, u^0) \cdot dp = x^0 \cdot dp$.

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NES

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This is the integrability problem. Converse theorems require additional domain and regularity conditions.

Reserve: pairwise consistency misses a longer cycle

Three goods; choose $x^A = e_1$, $x^B = e_2$, $x^C = e_3$, each with spending 2:

	p_1	p_2	p_3
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GARP rules out a revealed-preference chain followed by a strict reverse affordability comparison.

Reserve: genuine net complements need another good

Let $u(x) = \log(\min\{x_1, x_2\}) + x_3$, with $x_1, x_2 > 0$, $x_3 \geq 0$.

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Goods 1 and 2 are net complements:

$$\frac{\partial h_1}{\partial p_2} = -\frac{p_3}{(p_1 + p_2)^2} < 0.$$

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Those cost bounds make minimisers locally bounded. A convergent subsequence is target-feasible by continuity, giving $\liminf e(p^k, \bar{u}^k) \geq e(p, \bar{u})$.